

Linear Algebra - CS - 3A & 3B

Problem Sheet - Week 3 and 4

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Row Operations , (Reduced) Row Echolon Form

1. Enlist elementary row operations and define elementary matrices.
2. Show that matrix has **unique** RREF.
3. Every matrix can be written as product of sequence of elementary matrices (corresponding to elementary row operations). Show that this sequence is not unique. You may show this by given any particular example.
4. For finding determinant, argue that which method is more computationally expensive ?
 - (a) Adjoint Method
 - (b) Row Operation Method
5. Transform the matrices into Reduced Row Echelon Form (RREF). Mention the step where find matrix satisfy Row Echelon Form (REF):

(a)

(b)

(c)

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 7 \\ 3 & 6 & 10 \end{bmatrix}$$

$$B = \begin{bmatrix} 0 & 1 & 2 \\ 1 & 0 & 3 \\ 2 & 1 & 1 \end{bmatrix}$$

$$C = \begin{bmatrix} 1 & 2 & -1 & 4 \\ 2 & 4 & 1 & 9 \\ 3 & 6 & 0 & 13 \end{bmatrix}$$

(d)

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 1 & 2 & 3 & 14 \\ 2 & 3 & 4 & 20 \end{array} \right]$$

(e)

$$\begin{bmatrix} 0 & 1 & 2 \\ 1 & -1 & 3 \\ 2 & 1 & 1 \\ 0 & 1 & 1 \end{bmatrix}$$

(f)

$$\begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 2 & 1 & 3 \\ 3 & 1 & 2 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix}$$

6. In computer graphics transformation pipeline, the homogeneous coordinates transformation matrix is:

$$T = \begin{bmatrix} 1 & 2 & 1 & 4 \\ 2 & 4 & 1 & 5 \\ 3 & 6 & 2 & 9 \\ 0 & 0 & 1 & 2 \end{bmatrix}$$

Reduce T to REF to analyze the transformation's rank and determine if it preserves dimensions.

(**Fun Activity:** Load a small test image and convert in pixel RGB matrix. Apply transformation and see how it works!)

Computing Inverse Using Row Operations

7. Find the inverse of the following matrix using row operations:

(a)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 9 \end{bmatrix}$$

(b)

$$D = \begin{bmatrix} 2 & 1 \\ 5 & 3 \end{bmatrix}$$

(c)

$$E = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 1 & 1 \\ 1 & 1 & 0 \end{bmatrix}$$

Can you recognize matrix in Part (a)?

8. In 3D computer graphics, the transformation matrix for scaling by 2 in x , 3 in y , and rotating 90° around z -axis is:

$$M = \begin{bmatrix} 0 & -2 & 0 \\ 3 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Find M^{-1} using row operations for the inverse transformation.

9. In error-correcting codes (Hamming code), the parity check matrix is:

$$H = \begin{bmatrix} 1 & 0 & 1 & 1 \\ 1 & 1 & 0 & 1 \\ 0 & 1 & 1 & 1 \end{bmatrix}$$

Augment with I_3 and reduce to find the right inverse (if it exists) for syndrome decoding.

Compute determinant using row and column operations

10. Compute the determinant using row operations:

(a)

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 3 & 9 \end{bmatrix}$$

(b)

$$\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix}$$

(c)

$$\begin{bmatrix} 2 & 1 & 3 & 1 \\ 1 & 2 & 1 & 3 \\ 3 & 1 & 2 & 1 \\ 1 & 3 & 1 & 2 \end{bmatrix}$$

11. In computational geometry, the volume of a parallelepiped formed by vectors $\mathbf{u} = (1, 2, 1)$, $\mathbf{v} = (2, 1, 3)$, $\mathbf{w} = (3, 1, 2)$ is given by the absolute value of the determinant. Compute $\det[\mathbf{u} \ \mathbf{v} \ \mathbf{w}]$ using row operations to determine if the vectors are coplanar.
12. In graph theory, the Laplacian matrix of a small network is:

$$L = \begin{bmatrix} 2 & -1 & 0 & -1 \\ -1 & 3 & -1 & -1 \\ 0 & -1 & 2 & -1 \\ -1 & -1 & -1 & 3 \end{bmatrix}$$

The number of spanning trees is related to any cofactor. Compute $\det(L')$ where L' is L with row 1 and column 1 removed, using row/column operations for efficiency.

Definition 1. The Degree matrix D where D_{ii} = degree of vertex i (number of edges incident to it), and off-diagonal entries are zero.

Definition 2. Adjacency matrix is a square matrix A where $A_{ij} = 1$ if vertices i and j are connected by an edge, and 0 otherwise (for unweighted graphs).

Definition 3. A Laplacian matrix L is matrix obtained from Degree matrix and Adjacency matrix $L = D - A$.